

An Assessment of Potential Exposure and Risk from Estrogens in Drinking Water

Background Detection of estrogens in the environment has raised concerns in recent years because of their potential to affect both wildlife and humans. Objectives We compared exposures to prescribed and naturally occurring estrogens in drinking water to exposures to naturally occurring background levels of estrogens in the diet of children and adults and to four independently derived acceptable daily intakes (ADIs) to determine whether drinking water intakes are larger or smaller than dietary intake or ADIs. Methods We used the Pharmaceutical Assessment and Transport Evaluation (PhATE) model to predict concentrations of estrogens potentially present in drinking water. Predicted drinking water concentrations were combined with default water intake rates to estimate drinking water exposures. Predicted drinking water intakes were compared to dietary intakes and also to ADIs. We present comparisons for individual estrogens as well as combined estrogens. Results In the analysis we estimated that a child's exposures to individual prescribed estrogens in drinking water are 730–480,000 times lower (depending upon estrogen type) than exposure to background levels of naturally occurring estrogens in milk. A child's exposure to total estrogens in drinking water (prescribed and naturally occurring) is about 150 times lower than exposure from milk. Adult margins of exposure (MOEs) based on total dietary exposure are about 2 times smaller than those for children. Margins of safety (MOSs) for an adult's exposure to total prescribed estrogens in drinking water vary from about 135 to > 17,000, depending on ADI. MOSs for exposure to total estrogens in drinking water are about 2 times lower than MOSs for prescribed estrogens. Depending on the ADI that is used, MOSs for young children range from 28 to 5,120 for total estrogens (including both prescribed and naturally occurring sources) in drinking water. Conclusions The consistently large MOEs and MOSs strongly suggest that prescribed and total estrogens that may potentially be present in drinking water in the United States are not causing adverse effects in U.S. residents, including sensitive subpopulations.